Topic 6 — Joint Distributions, the Law of Large Numbers, and the Central Limit Theorem

From Single to Joint Random Variables

For a random variable X, we know how to calculate $\mathbb{E}(X)$ and $\mathrm{Var}(X)$. Now we consider how to characterise a **pair** of random variables $X, Y : \Omega \to \mathbb{R}$.

Definition

The **joint cumulative distribution function** (joint CDF) of $X, Y : \Omega \to \mathbb{R}$ is the function $F_{X,Y} : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$F_{X,Y}(x, y) := \mathbb{P}(X \leq x,\ Y \leq y).$$

(The comma inside $\mathbb{P}(\cdot)$ means "and".)

For the discrete case, if $X : \Omega \to \{x_1, x_2, \dots\}$ and $Y : \Omega \to \{y_1, y_2, \dots\}$, we consider the **joint probability mass function**

$$p_{X,Y}(x_i, y_j) = \mathbb{P}(X = x_i,\ Y = y_j), \qquad F_{X,Y}(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p_{X,Y}(x_i, y_j).$$

Example: A Joint PMF Table

Example

Let the joint distribution of $X : \Omega \to \{0, 1\}$ and $Y : \Omega \to \{0, 1, 2, 3\}$ be given by

$X \backslash Y$	0	1	2	3
0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0
1	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$

so, e.g., $p_{X,Y}(1, 2) = \mathbb{P}(X = 1, Y = 2) = \frac{2}{8}$.

Example

Summing rows and columns gives the **marginal** distributions:

$$p_X(0) = \tfrac{4}{8} = \tfrac{1}{2}, \qquad p_X(1) = \tfrac{4}{8} = \tfrac{1}{2},$$

$$p_Y(0) = \tfrac{1}{8}, \qquad p_Y(1) = \tfrac{3}{8}, \qquad p_Y(2) = \tfrac{3}{8}, \qquad p_Y(3) = \tfrac{1}{8}.$$

So we reconstructed p_X and p_Y from $p_{X,Y}$ (and hence F_X, F_Y) — **but the opposite process is not unique, and hence impossible in general.**

The Joint PDF (Continuous Case)

Definition

Let $X, Y : \Omega \to \mathbb{R}$ be continuous random variables. Their **joint probability density function** (joint PDF) is the function $f_{X,Y} : \mathbb{R}^2 \to \mathbb{R}$, $f_{X,Y} \geq 0$, such that

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \Big(\int_{-\infty}^{y} f_{X,Y}(u,v)\,\mathrm{d}v \Big) \mathrm{d}u, \qquad \int_{-\infty}^{\infty} \Big(\int_{-\infty}^{\infty} f_{X,Y}(u,v)\,\mathrm{d}v \Big) \mathrm{d}u = 1.$$

Remark

For the given joint PDF $f_{X,Y}$, the **marginal PDFs** are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}x.$$

This is the continuous analogue of $p_X(x_i) = \sum_j p_{X,Y}(x_i, y_j)$, $p_Y(y_j) = \sum_i p_{X,Y}(x_i, y_j)$ above.

Remark

Discrete: $\mathbb{E}(XY) = \sum_i \sum_j x_i y_j\, p_{X,Y}(x_i, y_j)$, Continuous: $\mathbb{E}(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x\, y\, f_{X,Y}(x,y)\, dy\, dx$.

Independence of Two Random Variables

Definition

Two random variables X and Y are **independent** if, for any $a, b \in \mathbb{R}$, the events $\{X \leq a\}$ and $\{Y \leq b\}$ are independent, i.e.

$$F_{X,Y}(x, y) = F_X(x) \cdot F_Y(y) \qquad \text{for all } x, y.$$

Then, in the discrete case, $p_{X,Y}(x_i, y_j) = p_X(x_i) \cdot p_Y(y_j)$, and in the continuous case, $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$. In both cases,

$$\mathbb{E}(XY) = \mathbb{E}(X)\,\mathbb{E}(Y).$$

Covariance and Correlation

Definition

The **covariance** of $X, Y : \Omega \to \mathbb{R}$ is

$$\operatorname{cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y))).$$

Remark

- $\operatorname{cov}(X,Y) = \operatorname{cov}(Y,X)$
- $\operatorname{cov}(X,X) = \operatorname{Var}(X)$
- $\operatorname{cov}(aX,bY) = ab\ \operatorname{cov}(X,Y)$
- $\operatorname{Var}(aX+bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab\ \operatorname{cov}(X,Y)$

In particular, if X and Y are independent, $\operatorname{cov}(X,Y) = 0$ and $\operatorname{Var}(aX+bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y)$.

Definition

The **correlation** of X, Y is

$$\operatorname{corr}(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sigma(X)\cdot\sigma(Y)}, \qquad -1 \le \operatorname{corr}(X,Y) \le 1.$$

If X and Y are independent, then $\operatorname{corr}(X,Y) = 0$, i.e. X and Y are **uncorrelated**. **The opposite is not necessarily true.**

Example: Covariance and Correlation

Example

Using the same joint table as above, with marginals $p_X(0) = p_X(1) = \frac{1}{2}$ and $p_Y(0) = \frac{1}{8}, p_Y(1) = \frac{3}{8}, p_Y(2) = \frac{3}{8}, p_Y(3) = \frac{1}{8}$:

$$\mathbb{E}(X) = \tfrac{1}{2}, \qquad \mathbb{E}(Y) = \tfrac{3}{2}, \qquad \mathbb{E}(XY) = 1.$$

$$\operatorname{cov}(X, Y) = 1 - \tfrac{1}{2} \cdot \tfrac{3}{2} = \tfrac{1}{4} \quad \Longrightarrow \quad X \text{ and } Y \text{ are not independent.}$$

Example

$$\sigma^2(X) = \tfrac{1}{4}, \qquad \sigma^2(Y) = \tfrac{3}{4}, \qquad \sigma(X) = \tfrac{1}{2}, \qquad \sigma(Y) = \tfrac{\sqrt{3}}{2},$$

$$\operatorname{corr}(X, Y) = \frac{\frac{1}{4}}{\frac{1}{2} \cdot \frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}.$$

Independent ⇒ Uncorrelated (but not conversely!)

Remark

For independent random variables X, Y, $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$, hence $\operatorname{cov}(X, Y) = 0$: **independent random variables are uncorrelated.** The converse is false, as the following example shows.

Example

Let $X \sim U(-1, 1)$ and $Y = X^2$. Then $XY = X^3$, so

$$\operatorname{cov}(X, Y) = \mathbb{E}(X^3) - \mathbb{E}(X)\mathbb{E}(X^2).$$

Since $f_X(x) = \frac{1}{2}$ on $(-1, 1)$, $\mathbb{E}(X) = 0$ and

$$\mathbb{E}(X^3) = \frac{1}{2}\int_{-1}^{1} x^3\, dx = \frac{1}{2}\left[\frac{x^4}{4}\right]_{-1}^{1} = 0.$$

Therefore $\operatorname{cov}(X, Y) = 0$: X and Y are **uncorrelated**. However, $Y = X^2$ is a deterministic function of X, so X and Y are clearly **not independent**.

Independence of Several Random Variables

Definition

Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$. They are called **independent** if, for any $a_1, \dots, a_n \in \mathbb{R}$, the events $\{X_1 \leq a_1\}, \dots, \{X_n \leq a_n\}$ are independent, i.e. their joint CDF factorises:

$$F_{X_1,\dots,X_n}(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n).$$

Definition

$X_1, X_2, \dots$ are **independent and identically distributed** (i.i.d.) if any finite group of them is independent and they all share the same distribution $F_{X_1} = F_{X_2} = \cdots = F_X$, i.e. $X_1 \sim X$, $X_2 \sim X, \dots$

The Sample Average

Remark

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}(X) = \mu$, $\mathrm{Var}(X) = \sigma^2 < \infty$. Consider the sample average $\bar{X}_n = \frac{X_1 + \ldots + X_n}{n}$.

$$\mathbb{E}(\bar{X}_n) = \frac{1}{n}(\underbrace{\mathbb{E}(X_1)}_{=\mu} + \cdots + \underbrace{\mathbb{E}(X_n)}_{=\mu}) = \frac{1}{n} \cdot n \cdot \mu = \mu.$$

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\mathrm{Var}(X_1 + \cdots + X_n) \overset{\text{independence}}{=} \frac{1}{n^2}(\underbrace{\mathrm{Var}(X_1)}_{=\sigma^2} + \cdots + \underbrace{\mathrm{Var}(X_n)}_{=\sigma^2}) = \frac{\sigma^2}{n}.$$

So $\mathbb{E}(\bar{X}_n) = \mu$, $\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.

The Law of Large Numbers

Definition

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}(X) = \mu$, $\mathrm{Var}(X) = \sigma^2 < \infty$. Then $\bar{X}_n \to \mu$ **stochastically** (in probability): for each $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0, \qquad \text{equivalently} \qquad \lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| \leq \varepsilon) = 1.$$

Remark

The bigger n you take, the smaller the chance of $\{|\bar{X}_n - \mu| > \varepsilon\}$. We can choose ε arbitrarily small: LLN states there is high probability (close to 1) of $\bar{X}_n$ being very close to μ, if n is large enough.

Example: LLN and Relative Frequency

Example

Let A be a random event with $\mathbb{P}(A) = p$. Let $X = 1$ if A holds, $X = 0$ otherwise (Bernoulli), and $X_1, \dots, X_n, \dots$ i.i.d. with $X_n \sim X$. Then $\mu = \mathbb{E}(X) = p$, and the sample average

$$\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$$

is the **frequency** of A among n trials. LLN says

$$\frac{\text{number of trials when } A \text{ happened}}{\text{number } n \text{ of all trials}} \to \mathbb{P}(A), \qquad n \to \infty,$$

matching our intuitive understanding of probability.

Example: Rolling a Die (LLN)

Example

Consider many rolls of a fair six-sided die. Let X_j be the score of the j-th roll, $S_n = X_1 + ... + X_n$. All X_j are i.i.d. with

$$\mathbb{E}(X) = 1 \cdot \tfrac{1}{6} + \cdots + 6 \cdot \tfrac{1}{6} = \frac{7 \cdot 6}{2} \cdot \tfrac{1}{6} = \frac{7}{2}.$$

By LLN, for any small $\varepsilon > 0$,

$$\lim_{n \to \infty} \mathbb{P}\left(\left| \tfrac{S_n}{n} - \tfrac{7}{2} \right| \leq \varepsilon \right) = 1.$$

The Central Limit Theorem: Setup

By LLN, $\bar{X}_n \to \mathbb{E}(X) = \mathbb{E}(\bar{X}_n)$, i.e. $\bar{X}_n - \mathbb{E}(\bar{X}_n) \to 0$. The **CLT** tells us *how fast* this happens. Recall $\sigma(\bar{X}_n) = \dfrac{\sigma(X)}{\sqrt{n}}$. Define

$$Z_n := \frac{\bar{X}_n - \mathbb{E}(\bar{X}_n)}{\sigma(\bar{X}_n)} = \frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu).$$

One checks $\mathbb{E}(Z_n) = 0$ and $\mathrm{Var}(Z_n) = 1$: numerator and denominator both $\to 0$ as $n \to \infty$, but their ratio is standardised at every n.

The Central Limit Theorem

Definition

Let $X_1, \dots, X_n, \dots$ be i.i.d. with $X_n \sim X$, $\mu := \mathbb{E}(X)$, $\sigma^2 := \mathrm{Var}(X) < \infty$, and Z_n as above. Then

$$Z_n \to Z \sim \mathcal{N}(0,1), \qquad n \to \infty,$$

in the sense that $\lim_{n\to\infty} \mathbb{P}(Z_n \le z) = \Phi(z)$ for all $z \in \mathbb{R}$. As a corollary,

$$\lim_{n\to\infty} \mathbb{P}(a \le Z_n \le b) = \Phi(b) - \Phi(a), \qquad \lim_{n\to\infty} \mathbb{P}(Z_n \ge c) = 1 - \Phi(c).$$

Remark

The CLT shows that $\bar{X}_n$ fluctuates around μ with standard deviation $\sigma(\bar{X}_n) = \dfrac{\sigma}{\sqrt{n}}$, significantly less than σ itself — regardless of the distribution of X_n.

Example: Teacher Salaries (CLT)

Example

The average teacher's salary in New Jersey in 2023 is \$63,178, normally distributed with standard deviation \$7,500: $X \sim \mathcal{N}(63178, 7500^2)$.
Probability a randomly selected teacher makes less than \$60,000:

$$\mathbb{P}(X < 60000) = \mathbb{P}\Big(\underbrace{\tfrac{X-63178}{7500}}_{=:Z} < \tfrac{60000-63178}{7500}\Big) = \mathbb{P}(Z < -0.42) = \Phi(-0.42) \approx 0.3372.$$

```
from scipy.stats import norm
norm.cdf(-0.42)
```

```
0.3372427268482495
```

Roughly each third teacher earns less than \$60,000.

Example

Now take the average of 100 teachers' salaries, $\bar{X}_{100} = \frac{X_1+\cdots+X_{100}}{100}$. Then $\mathbb{E}(\bar{X}_{100}) = 63178$ and $\sigma(\bar{X}_{100}) = \dfrac{7500}{\sqrt{100}} = 750$, so

$$\mathbb{P}(\bar{X}_{100} < 60000) = \mathbb{P}\Big(Z_{100} < \tfrac{60000-63178}{750}\Big) = \mathbb{P}(Z_{100} < -4.2) \underset{\text{CLT}}{\approx} \Phi(-4.2) \approx 0.0000133.$$

```
norm.cdf(-4.2)
```

```
1.3345749015906314e-05
```

This is very unlikely — averaging shrinks the spread dramatically.

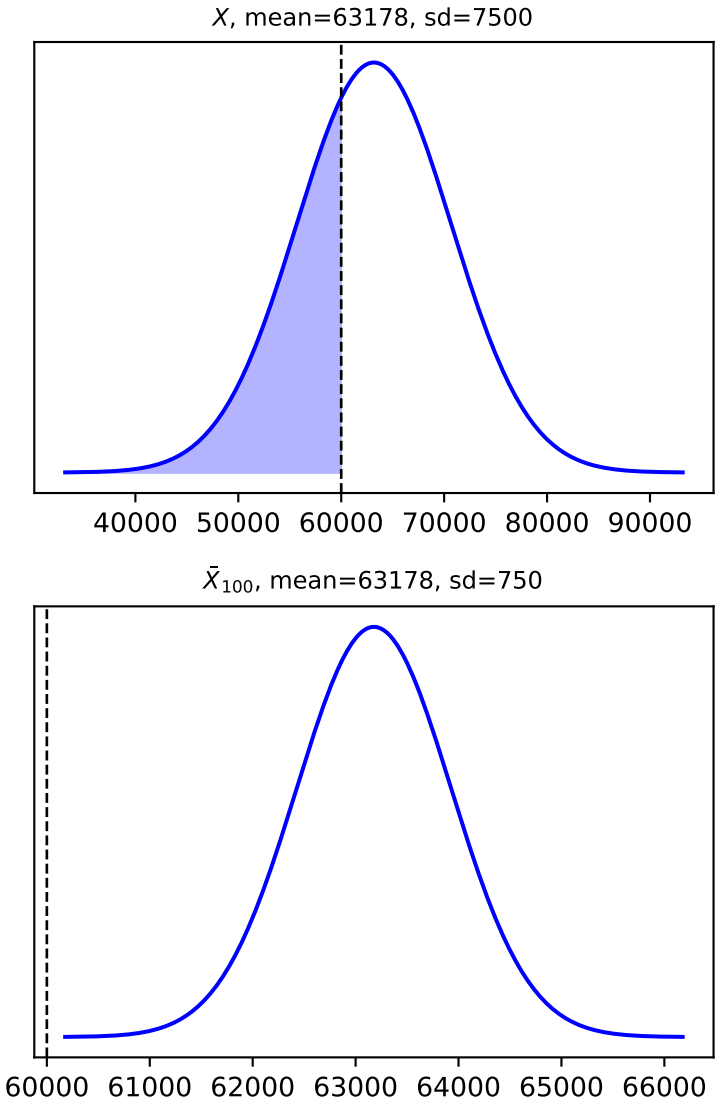


Figure 1: Salary distribution X (wide) vs. sample-average distribution $\bar{X}_{100}$ (narrow), both with mean \$63,178. The shaded left tail up to \$60,000 is much larger for X (≈ 0.34) than for $\bar{X}_{100}$ (≈ 0.0000133).